

WEATHER REGRESSION

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Why are we doing this?

Does climate
(temperature, humidity
, and wind speed?) vary
along an urbanization
gradient in St. Louis?

Backcast the growing
degree hours from
earliest to latest date.

METHOD

- WHERE DID THE DATA COME FROM?
- <http://agebb.missouri.edu/weather/history/index.asp>
- Select into Saint Louis Science Center/County
- GIS data: <https://docs.google.com/spreadsheets/d/1Itrw-fDMKWzntzephPAaHWW6bYDhvjp8LzaA6TRBHM/edit?gid=0#gid=0>
- Working Folder > Weather

**Missouri Historical Agricultural Weather Database**

This model is used to query the Missouri Agricultural Weather Database to obtain hourly and daily weather data from the Commercial Agriculture Automated Weather Station Network.

Weather Location: Albany (Gentry County) [View Map](#)

	<u>Starting Period:</u>		<u>Ending Period:</u>
Month:	January ▾	Month:	January ▾
Day:	1 ▾	Day:	1 ▾
Year:	2025 ▾	Year:	2025 ▾



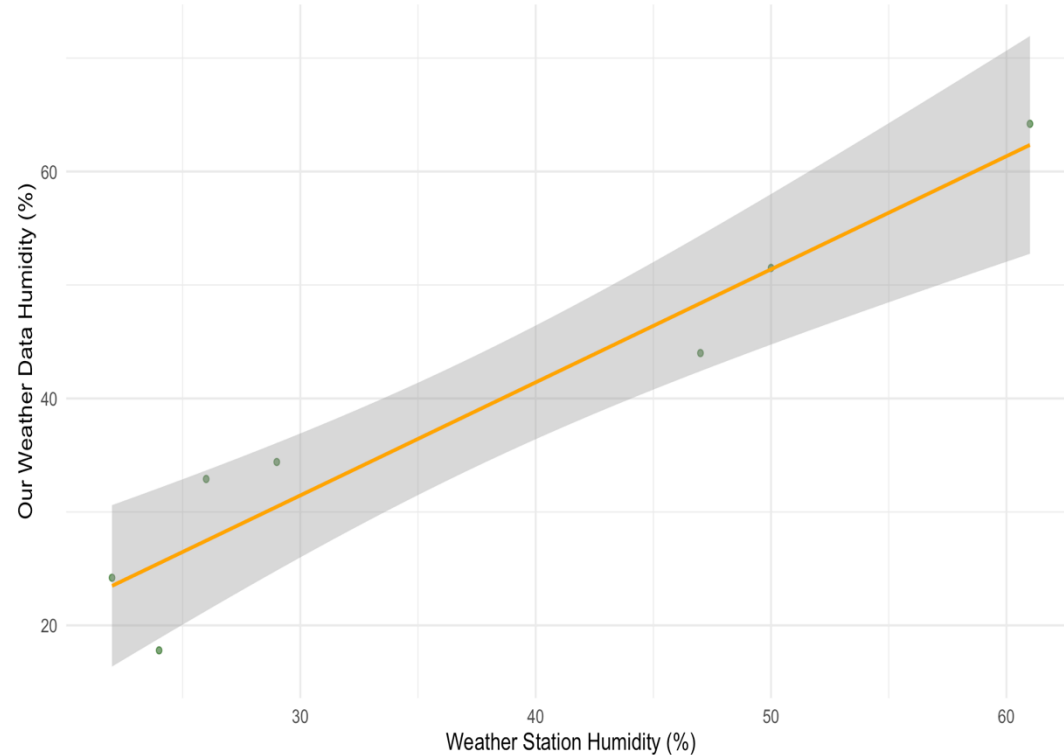
Method contd...

- GDH - calculates hour by hour instead of daily
 - We take hourly temperature data and sums all the degrees above a base temperature (in our case $> 50^{\circ}\text{F}$), hour by hour.
 - GDD - measures how many degrees above a base temperature (50°F) the day was.
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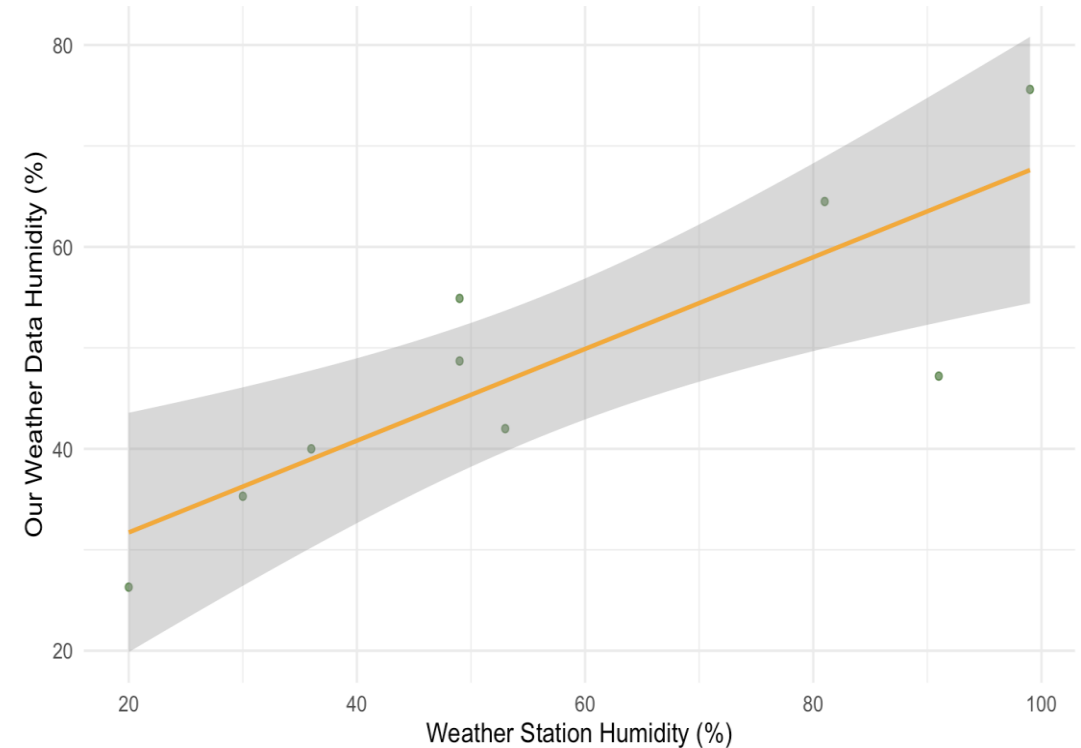
Result: Humidity

Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
McKinley	Humidity	0.996348314606742	1.56368378812198	55.5513691030597	0.000685884819849864	0.917425483947524

Humidity Comparison at McKinley



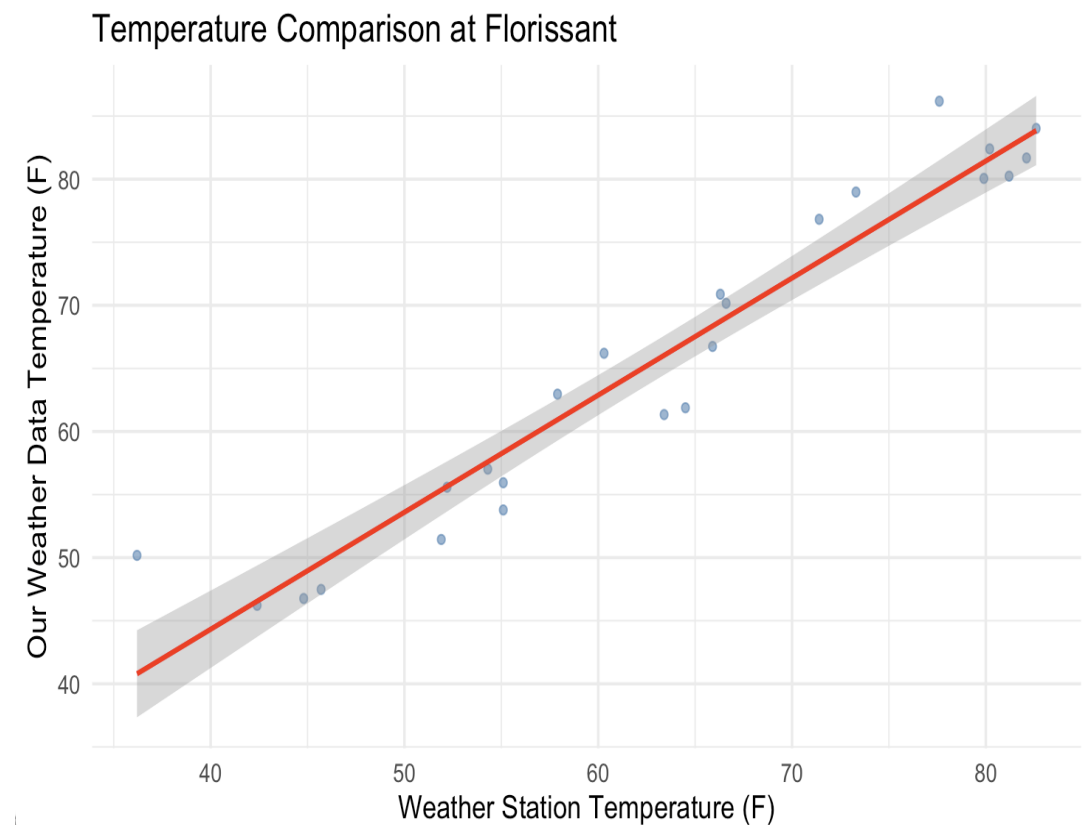
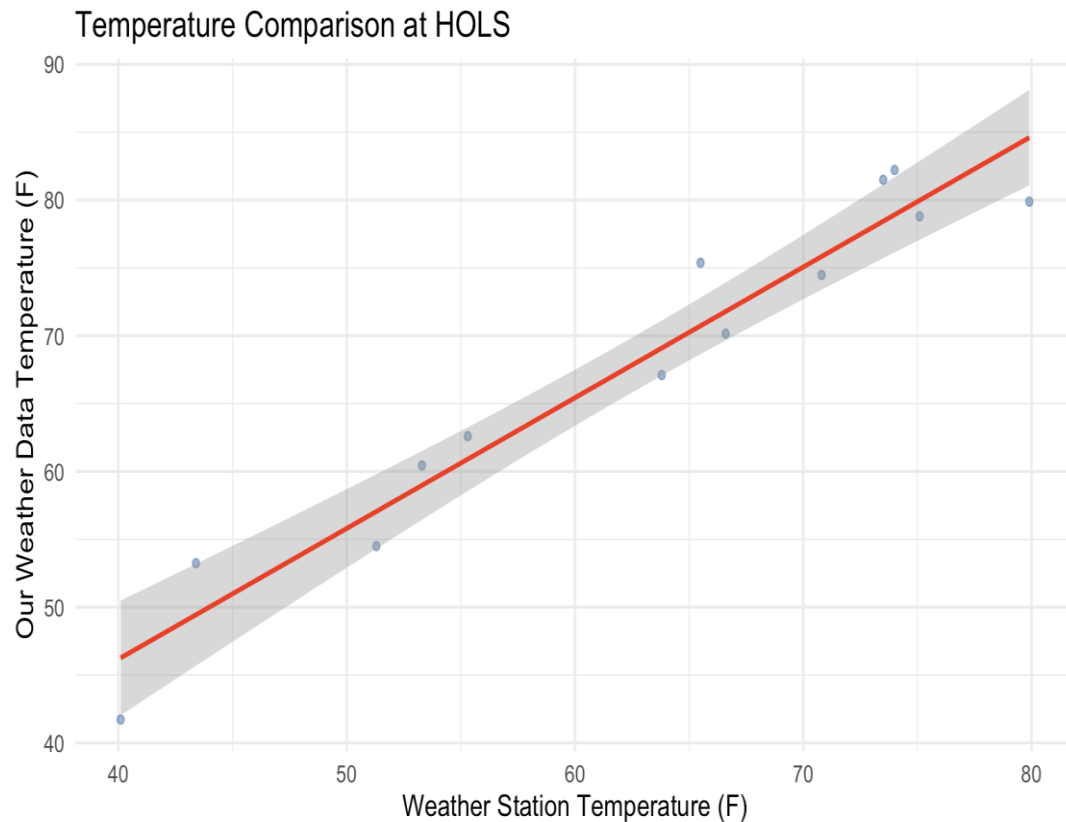
Humidity Comparison at Virginia



Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
13th Street	Humidity	0.38612100993252	27.0316324209568	9.78578330094363	0.0166510243957224	0.582980437999191
Carondelet	Humidity	0.98110829639012	3.8104749841672	13.3211996866016	0.00446361637460101	0.57120559257742
COLA	Humidity	0.872151057401812	5.47310574018128	74.4446499914583	2.0521764484363E-07	0.82309622513315
EarthDance	Humidity	0.916188113932783	4.32682742534249	46.1010274787937	1.33123110976711E-06	0.697432842380305
Emmanuel	Humidity	0.262677705100046	32.6978832275349	5.28408045702059	0.0612083798775845	0.468277453102792
Ferguson	Humidity	0.780173820188058	9.80273485005782	14.8120969888079	0.00117665112933606	0.451421833656662
Florissant	Humidity	0.916746553099128	3.58300762141231	84.6752415555995	5.36176057667321E-09	0.793766579018867
HOLS	Humidity	0.637391860961322	20.4397777088102	70.4042829581106	4.13921036615417E-06	0.86487197478712
Holy Cross	Humidity	0.392687180764918	26.4888127569453	6.43252046743403	0.044304968092185	0.517394721712583
Kellogg	Humidity	0.634423748865106	14.8958566043965	43.1123270723623	6.3436762979487E-05	0.811719791784389
McKinley	Humidity	0.996348314606742	1.56368378812198	55.5513691030597	0.000685884819849864	0.917425483947524
Rustic Roots	Humidity	0.68671655556953	16.9291430847273	33.8354892419057	5.35478235526702E-06	0.585029878460674
SLU	Humidity	0.343940294413459	31.889512663436	9.9709974481553	0.0159797293453033	0.587531609654394
Thies	Humidity	0.759281914893617	13.1892021276596	16.6281738438308	0.0151261403999705	0.806090445509976
Virginia	Humidity	0.454385996473932	22.6302126434714	16.5592390439759	0.00475339780381625	0.702876651196852

Result: Temperature

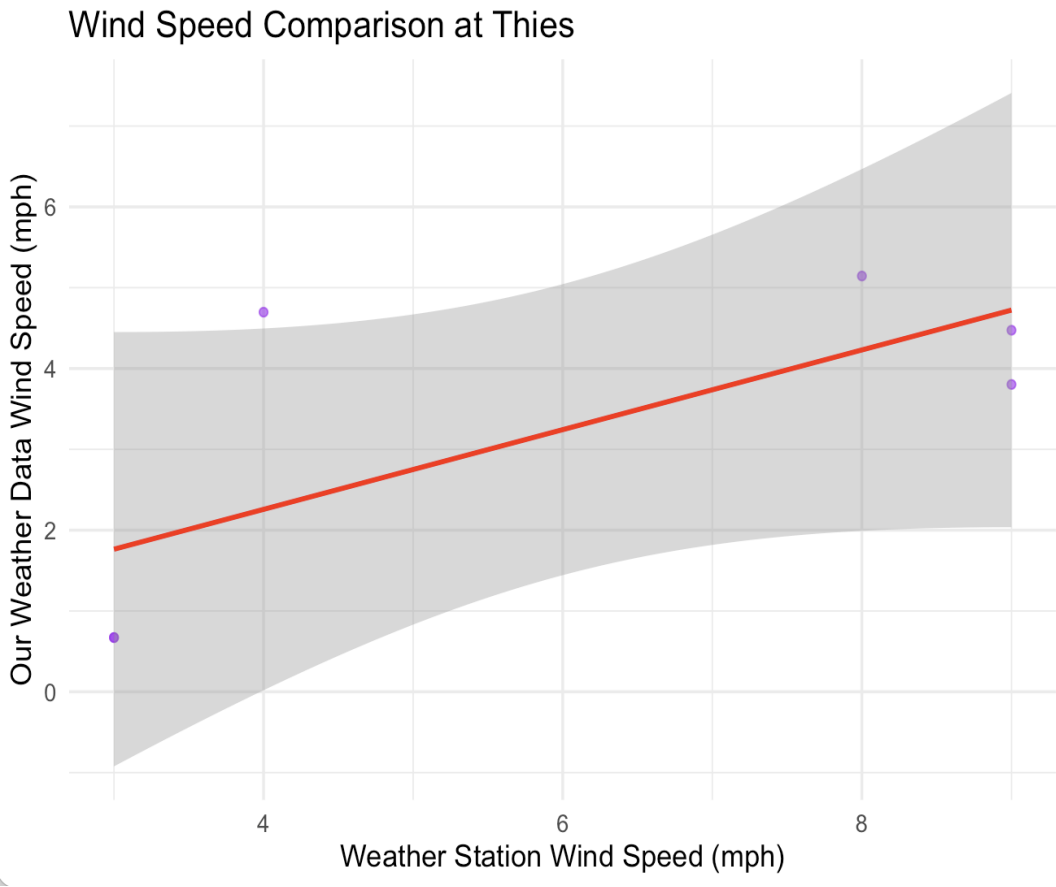
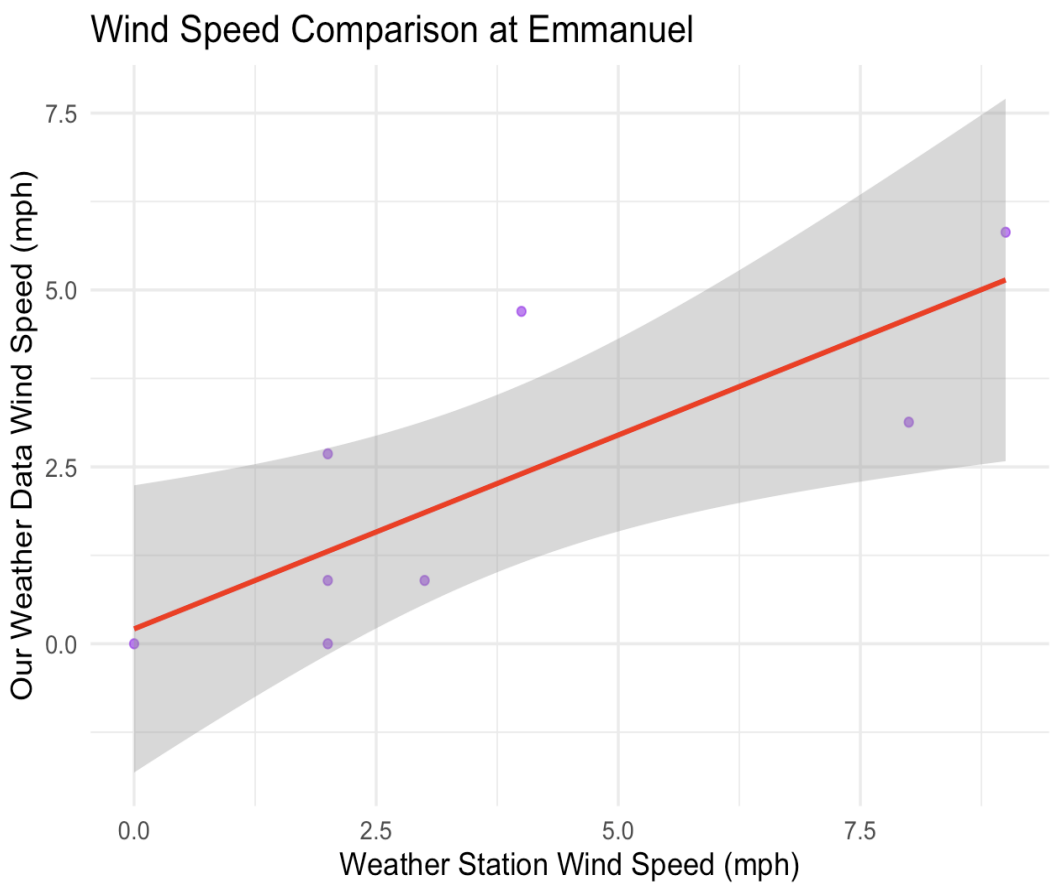
Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
HOLS	Temperature	0.963404723455153	7.62748628618023	164.152642734931	5.91731038318161E-08	0.937197636140455



Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
13th Street	Temperature	0.848773769689056	14.7994413612993	41.1999186829071	0.000360806753501307	0.854771539220825
Carondelet	Temperature	0.821542201507707	14.3902280708814	58.8429491417932	1.69788059674603E-05	0.854741841762133
COLA	Temperature	1.14025146022592	-5.1663941067967	124.654767826074	5.80759055257093E-09	0.886246301868809
EarthDance	Temperature	0.966318985096722	7.6006710724317	103.036176277799	2.45885850361189E-09	0.837446183674932
Emmanuel	Temperature	0.478615844794387	33.7044665758408	3.82629216119404	0.0982260001850667	0.389393282677348
Ferguson	Temperature	0.935954212020824	7.57262182239199	165.888989922798	7.77252489423283E-11	0.89723563307943
Florissant	Temperature	0.927969649898918	7.2029439986552	278.939877805397	5.54926498960664E-14	0.926895697039439
HOLS	Temperature	0.963404723455153	7.62748628618023	164.152642734931	5.91731038318161E-08	0.937197636140455
Holy Cross	Temperature	1.04587944681022	0.122734426492625	67.4789441150968	0.000175624018630234	0.918343954553815
Kellogg	Temperature	0.81641466657656	19.5418079500273	63.77987673927	1.19539242476493E-05	0.864461687360377
McKinley	Temperature	0.727576441867561	27.4280651815685	6.14286343346449	0.0559452393277841	0.551282304601895
Rustic Roots	Temperature	0.971759972138684	3.99939568105806	238.63683723865	1.23313543041455E-13	0.912091889495589
SLU	Temperature	0.817087074669188	10.8915533435727	20.6803611180369	0.00264374863252221	0.747113125795216
Thies	Temperature	1.1752360526247	-9.09728977584807	52.1321413806904	0.00195140479092833	0.928739579470667
Virginia	Temperature	0.896098751255921	10.9169490455002	36.2077701567765	0.000532998690418013	0.837992102471362

Result: Wind Speed

Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
Emmanuel	Wind Speed	0.547969834532374	0.210014870503597	9.93674204008171	0.0197579932197217	0.623511506623518



Location	Variable	Slope	Intercept	F_Statistic	P_Value	Multiple_R2
13th Street	Wind Speed	0.186411666666667	0.658654555555555	3.84615384615385	0.0906644213959706	0.354609929078014
Carondelet	Wind Speed	1.11216547181628	3.5536523862213	0.381221751932361	0.550749946739124	0.0367222433969682
COLA	Wind Speed	0.23737752273936	1.69956746169817	0.356093005374492	0.559033951505148	0.0217712754052868
EarthDance	Wind Speed	-0.00611575187269994	2.43385211274465	0.000822068672784232	0.977425418464373	4.32649003192131E-05
Emmanuel	Wind Speed	0.547969834532374	0.210014870503597	9.93674204008171	0.0197579932197217	0.623511506623518
Ferguson	Wind Speed	0.111901998745236	1.42727255557327	0.801334848790762	0.382509395753793	0.0426211678710834
Florissant	Wind Speed	0.00986333717986695	2.94355397471104	0.00316764083002185	0.955625352292927	0.000143962945778036
HOLS	Wind Speed	0.103679673427992	2.43998881338743	0.443026467552087	0.519378488952876	0.0387158474909007
Holy Cross	Wind Speed	0.552904037735849	0.316548113207547	3.86586403684421	0.0968617632409068	0.39184241972189
Kellogg	Wind Speed	0.108878063615561	1.74425012585812	0.784186569513505	0.396658298890812	0.0727163392861146
McKinley	Wind Speed	-0.14249001369863	1.88914179452055	0.742798742678507	0.428155407778807	0.129344379972483
Rustic Roots	Wind Speed	0.0641287210052941	1.82854538757651	0.184309779322642	0.671524844085322	0.00762104773733196
SLU	Wind Speed	0.209645584541063	1.28344882769726	2.34594527208128	0.169468974313473	0.2510120917452
Thies	Wind Speed	0.493143590909091	0.284701454545454	4.24163191705173	0.108511324304431	0.514659227655618
Virginia	Wind Speed	0.169195836581709	1.08728028185907	1.18537086631474	0.312322738161659	0.144815779965805

GDH

- GDH50 ~ `PC1 500m` + (1 | YEAR)

Location	YEAR	GDH50
McKinley	2024	1422
Emmanuel	2024	1409
Kellogg	2024	1373
13th Street	2024	1269
Carondelet	2024	1203
McKinley	2022	1200
McKinley	2023	1200
Virginia	2024	1199
Emmanuel	2023	1195
Kellogg	2023	1192
Emmanuel	2022	1191

REML criterion at convergence: 541

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.5898	-0.7350	-0.0643	0.6093	2.2267

Random effects:

Groups	Name	Variance	Std.Dev.
YEAR	(Intercept)	9330	96.59
Residual		12471	111.67

Number of obs: 45, groups: YEAR, 3

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1067.32	58.20	2.00	18.338	0.00296 **
`PC1 500m`	31.17	10.99	41.00	2.837	0.00705 **

GDD

- GDD50 ~ `PC1 500m` + (1 | YEAR)

Location	YEAR	GDD50
13th Street	2022	13284.99
13th Street	2023	51667.23
13th Street	2024	109242.50
COLA	2022	125449.28
COLA	2023	161305.48
COLA	2024	236951.34
Carondelet	2022	230443.57
Carondelet	2023	264348.02
Carondelet	2024	358739.45
EarthDance	2022	332420.38

REML criterion at convergence: 1262.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.73906	-0.73240	-0.04612	0.86703	1.86870

Random effects:

Groups	Name	Variance	Std.Dev.
YEAR	(Intercept)	5.128e-07	7.161e-04
Residual		2.699e+11	5.195e+05

Number of obs: 45, groups: YEAR, 3

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.529e+05	7.748e+04	2.398e+15	11.008	<2e-16 ***
`PC1 500m`	4.036e+04	5.112e+04	1.402e+36	0.789	0.43

THANK YOU

